

JEE MAIN REVISION MASTER: LIQUID SOLUTION

2026 4 QUESTIONS 6.67% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Colligative Properties & Van 't Hoff Factor	17 PYQs	Colligative
Very High	Raoult's Law & Vapor Pressure (Ideal & Non-ideal)	16 PYQs	Vapor Pressure

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Colligative Properties & Van 't Hoff Factor	$\frac{P^\circ - P}{P^\circ} = i \cdot x_B$, $\Delta T_b = i K_b m$, $\Delta T_f = i K_f m$, $\Pi = i C R T$, $i = 1 + (n - 1)\alpha$
Raoult's Law & Vapor Pressure (Ideal & Non-ideal)	$P_{\text{total}} = P_A^\circ x_A + P_B^\circ x_B$, $y_A = \frac{P_A}{P_{\text{total}}}$, $\frac{P_A}{P_{\text{total}}} = \frac{P_A^\circ x_A}{P_{\text{total}}}$, $P = K_H \cdot x$ (Henry)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Colligative Properties & Van 't Hoff Factor (Very High Yield — 17 PYQs)

Core Principle: Physical properties depending solely on solute particle count, cryoscopic/ebullioscopic constants, and Van 't Hoff factor i .

■■ **Caution / JEE Pitfall:** When determining relative boiling/freezing points of salt solutions, compare effective molality $i \times m$, NOT just nominal molality m !

■ **Speed Trick:** For dimerization/association of n molecules: $i = 1 - \left(1 - \frac{1}{n}\right)\alpha$ implies $i < 1$ (e.g. benzoic acid in benzene, $i \approx 0.5$).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The relative lowering of vapour pressure produced by dissolving 18 g of urea (Molar mass = 60 g mol⁻¹) in 100 g of water is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "When 0.0106 mole of acetic acid was dissolved in 1 kg of water, the freezing point depression for this strength of acid was 0.0205 K. ..." [Ans: D]
- [2026 Shift] Q#4 (MCQ): "Which of the following is CORRECT with respect to the property mentioned against it?..." [Ans: D]

3.2 Raoult's Law & Vapor Pressure (Ideal & Non-ideal) (Very High Yield — 16 PYQs)

Core Principle: Partial pressures of volatile mixtures, Dalton's law in vapor phase (y_A), and thermodynamic criteria for deviations (ΔH_{mix} , ΔV_{mix}).

■ **Caution / JEE Pitfall:** Mixture of Chloroform and Acetone exhibits **NEGATIVE** deviation from Raoult's law ($\Delta H_{\text{mix}} < 0$, $\Delta V_{\text{mix}} < 0$) due to intermolecular hydrogen bonding.

■ **Speed Trick:** Higher Henry's law constant K_H at a given pressure signifies **LOWER** solubility of the gas in liquid.

Representative Source PYQs:

- **[2026 Shift] Q#3 (MCQ):** "Match List-I (Laws) with the List-II (Mathematical expressions): List-I List-II (a) Henry's law (i) $p_1 = p_1^0 x_1$ (b) Raoult's law ..." [Ans: N/A]
- **[2025 Shift] Q#2 (MCQ):** "Variation of solubility with temperature t for a gas in liquid is shown by the following graphs. The correct representation is..." [Ans: N/A]
- **[2025 Shift] Q#4 (MCQ):** "If N_2 gas is bubbled through water at 293 K, how many moles of N_2 gas would dissolve in 1 litre of water? Assume that N_2 obeys..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The relative lowering of vapour pressure produced by dissolving 18 g of urea (Molar mass = 60 g mol⁻¹) in 100 g of water is

- (A) 0.025
- (B) 0.512
- (C) 0.051
- (D) 0.250

Correct Answer: None

Solution Analysis: For a non-volatile solute like urea, the relative lowering of vapour pressure is equal to the mole fraction of the solute. $\frac{p^0 - p}{p^0} = x_{\text{solute}}$ Moles of urea: $n_{\text{urea}} = (18 / 60) = 0.3 \text{ mol}$ Moles of water: $n_{\text{water}} = (100 / 18) = 5.56 \text{ mol}$ Now, mole fraction of urea is $x_{\text{urea}} = \frac{n_{\text{urea}}}{n_{\text{urea}} + n_{\text{water}}} = \frac{0.3}{0.3 + 5.56} = 0.051$ So, the relative lowering of vapour pressure is 0.051 Correct option: Option C.

Benchmark Problem 2: KCET 2026 (Q#3)

Match List-I (Laws) with the List-II (Mathematical expressions): List-I List-II (a) Henry's law (i) $p_1 = p_1^0 x_1$ (b) Raoult's law (ii) $p = K_H \chi$ (c) First law of thermodynamics (iii) $\Delta_m = v_+ \lambda_+ + v_- \lambda_-$ (d) Kohlrausch's law (iv) $\Delta U = q + w$ Codes:

- (A) a-i, b-ii, c-iii, d-iv
- (B) a-ii, b-i, c-iii, d-iv
- (C) a-ii, b-i, c-iv, d-iii
- (D) a-i, b-ii, c-iv, d-iii

Correct Answer: None

Solution Analysis: Henry's law states that the partial pressure of a gas in vapour phase is directly proportional to its mole fraction in the solution. So, $p = K_H \chi$ Hence, (a) Henry's law matches with (ii). Raoult's law states that the partial vapour pressure of a volatile component is equal to the product of its mole fraction and vapour pressure in pure state. So, $p_1 = p_1^0 x_1$ Hence, (b) Raoult's law matches with (i). First law of thermodynamics is based on conservation of energy. It is written as $\Delta U = q + w$ Hence, (c) First law of thermodynamics matches with (iv). Kohlrausch's law says that